MANDOROS

Meta-Analysis Network for Data, Outcomes & Research Optimization Synthesis

— From Raw Trials to Global Clinical Verdicts —

PRISMA 2020 Compliant | Cochrane Handbook Standards | metafor-Grade REML | Network Meta-Analysis & SUCRA | In-Silico Virtual Trials

📌 **الهوية الرسمية لمنصة MANDOROS والشعار المعتمد**

تم اعتماد **حرف الـ M الانسيابي ثلاثي الأبعاد المضيء (M-Ribbon Emblem)** رسميًا ليكون هوية المنصة والعلامة التجارية البصرية لـ MANDOROS. يرمز الشعار إلى ربط أذرع التجارب السريرية غير المتجانسة ودمجها في نسيج علمي موحد ينقل الطب القائم على الدليل من مرحلة البيانات الورقية الخام إلى أحكام سريرية قطعية واستكشافات افتراضية تنبؤية.

عند مشاركة رابط المنصة (**https://mandoros.onrender.com**) عبر واتساب أو لينكد إن أو إكس/تويتر، تظهر بطاقة المعاينة الاجتماعية الفخمة (OpenGraph Social Card) تلقائيًا بالشعار المضيء والعنوان الميتاليك والوصف الأكاديمي المعتمد.

🧬 The 8-Node MANDOROS Architectural Pipeline (M-A-N-D-O-R-O-S)

M	A	N	D	O	R	O	S
Meta-Analysis DL & REML Random-Effects	**AI Automation** Verbatim Quote Provenance	**Network NMA** Bucher & SUCRA Rankings	**Data Schema** 88 Atomic Value/Unit Fields	**Outcomes** Continuous, Binary & Survival	**Research** CONSORT, PRISMA & Cochrane	**Optimization** Scale-Free SMD Standardization	**Synthesis** In-Silico Virtual Trial Protocol

🏛️ The Four Academic & Methodological Pillars

Pillar I · Atomic Curation — STAGE-1 ENGINE

Parses unstructured RCT PDF manuscripts into a 7-sheet canonical database. Deconstructs criteria with 4 relational operators (<, ≤, >, ≥) and enforces zero-imputation guardrails with verbatim citation tags.

Pillar II · metafor-Grade Pooling — CLASSICAL & REML

Executes Restricted Maximum Likelihood (REML) and DerSimonian-Laird random-effects pooling. Computes Cochran's Q, τ^2, Higgins' I^2, 95% Confidence Intervals, and 95% Prediction Intervals (PI).

Pillar III · Network Meta & SUCRA — MULTI-ARM NMA

Reconciles disparate active ingredients and comparator arms. Generates Monte Carlo SUCRA surface rankings, Bucher indirect comparisons, and node-splitting inconsistency audits.

Pillar IV · In-Silico Synthesis — EMERGENT DISCOVERY

Reconstructs missing trial cells via surrogate regression, performs dose-duration meta-regression, identifies therapeutic sweet-spots, and synthesizes de-novo virtual clinical trial protocols.

📑 Stage-1 Single-Study Extraction Specification (7 Canonical Sheets)

RELATIONAL SHEET	TARGET CLINICAL DOMAIN	ATOMIC COLUMNS EXTRACTED	VERIFICATION & EPISTEMIC STANDARD
1. Master_Extraction_Sheet	Consolidated Overview	Study ID, Title, DOI, Phase, Total N, Primary Endpoint, Safety Verdict	Page & quote coordinate provenance
2. Study_Characteristics	Design Rigor & Bias	Allocation, Masking, Setting, Centers, Funding Source, Risk of Bias	CONSORT Item 1-10 checklist audit
3. Population_Baseline	Demographics & Labs	Mean/SD Age, % Female, BMI, Disease Duration, Baseline Biomarkers	Cross-arm arithmetic conservation
4. Intervention_Arms	Regimens & Doses	Arm Name, Active Ingredient, Dosage, Route, Regimen Frequency	WHO ATC / RxNorm standardized
5. Outcomes	Efficacy & Safety Endpoints	Timepoint (Wks), Metric, Central Tendency, Dispersion, p-value	Value / Operator / Unit split
6. Variables_Matrix	Trajectory Grid	Normalized baseline vs multi-timepoint delta curves across all arms	Scale normalization & CFB audit
7. Eligibility_Criteria	Criteria Matrix	Category, Atomic Variable, Operator (<, ≤, >, ≥), Threshold, Unit	4-Operator mathematical validation

🚀 Phase-2 Multi-Trial Evidence Synthesis & In-Silico Discovery Lab

1. Missing-Data Matrix Reconstruction
When pooling heterogeneous trials, incomplete studies that omitted a specific follow-up timepoint or secondary biomarker are scientifically reconstructed using cross-study surrogate regression and longitudinal trajectory splines.

2. Dose–Duration Virtual Trial Simulator
Multivariate Weighted Least Squares (WLS) meta-regression models evaluate prospective target dosages (0–3,000 mg/day) and durations (4–156 weeks) with automated Extrapolation Risk Ratings (Interpolation vs Extrapolation).

3. SUCRA Treatment Hierarchy & Subgroups
Surface Under the Cumulative Ranking (SUCRA) scores quantify probability of superior efficacy across drug classes. Cochran's Q_between subgroup meta-analysis identifies significant interaction effects.

4. De-Novo Virtual Protocol Synthesis
Synthesizes the optimal next-generation trial protocol: pinpointing the therapeutic sweet-spot dose, duration saturation curve, super-responder phenotype, and statistical sample size for 90% power.

📦 Universal Export Center & Institutional Compliance

Universal Data Artifacts: Instant generation of 7-sheet single-study Excels, 10-sheet Master Cohort workbooks, standalone interactive HTML Forest/Funnel plot dashboards, structured JSON protocol trees, and individual per-study Excels packaged into an instant ZIP archive.